
R Tutorial 11

Instructions:

- Answer all questions.
- Ensure that your findings and results are clearly stated and thoroughly discussed. Please support your arguments using suitable R code with the relevant outputs, interpretations, plots and graphs whenever possible. You should support your argument using appropriate theory that is appropriately referenced.
- The R commands that you use in obtaining your results for all questions must be documented in a R script file. These scripts must be clearly commented. Ensure that any output is clearly stated and interpreted separately from the code as additional comments.
- Include the task name, your name and surname, and your student number in your R script file.
- You MUST label each answer by question number and, where a question has multiple parts, label each part of the question CLEARLY.
- On completion of your assignment, please submit onto RUconnected. If there are any issues uploading onto RUconnected, you may email your submission to: s.izally@ru.ac.za. Please submit your R script file and any other saved data files and plots mentioned in the questions below. Your student number should be included in the name of each file that you submit.
- Each student must complete an individual assignment. You will be assessed based on the quality and/or correctness of the R code, its outputs, and your explanations and interpretations. Acknowledge any help you may have received. Feel free to note any help you may have given to other students in the course.
- This assignment must be submitted by Tuesday, 8 October 2024 by 17:00. Late submissions will be penalized.
- Please note the Rhodes University and the Rhodes University Department of Statistics plagiarism policies.

Questions:

1. An urn contains ten marbles, of which five are green, two are blue, and three are red. Three marbles are to be drawn from the urn, one at a time without replacement.
 - (a) What is the probability that all three marbles drawn will be green?
 - (b) Determine the expected number of green marbles selected in a sample of size 5.
 - (c) Determine the variance of the number of green marbles selected in a sample of size 5.
2. A warehouse contains ten printing machines, four of which are defective. A company selects five of the machines at random, thinking all are in working condition. What is the probability that all five of the machines are non-defective?
3. A corporation is sampling without replacement for $n = 3$ firms to determine the one from which to purchase certain supplies. The sample is to be selected from a pool of six firms, of which four are local and two are not local. Let Y denote the number of non-local firms among the three selected. Determine:
 - (a) $P(Y = 1)$,
 - (b) $P(Y \geq 1)$,
 - (c) $P(Y \leq 1)$.